

**Project Interim Report  
On  
File Compression Simulator  
(Using Core Java & OOP Concepts)**

**RU-100-01-00012  
Object Oriented Programming With Java**

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## ABSTRACT

The Customer Feedback Analysis System is a Java-based project developed using Object-Oriented Programming (OOP) concepts. In the modern digital landscape, businesses receive extensive customer feedback via ratings and reviews. Manually analyzing this data is time-consuming and inefficient. This project automates the analysis process by taking multiple feedback entries, calculating the average rating, and categorizing sentiment into Excellent, Good, Average, or Poor. The system utilizes arrays, classes, and switch-case logic to ensure efficient operations. This implementation demonstrates the practical application of OOP in real-world scenarios while enhancing core programming skills.

## INTRODUCTION

Customer feedback is vital for businesses to improve their services, yet handling large volumes manually is prone to errors. This system is designed to solve these challenges by automating feedback processing. It stores feedback data using objects and processes it through Java programming concepts. The primary focus is calculating average ratings and determining overall sentiment. Key concepts applied include classes, objects, arrays, and decision-making structures like switch-case.

## OBJECTIVES

To analyze customer feedback programmatically.

- To calculate average ratings accurately.
- To categorize feedback into distinct sentiment levels.
- To apply Object-Oriented Programming concepts effectively.
- To improve problem-solving and coding skills.

## 5. SCOPE

This project is suitable for small businesses requiring quick feedback analysis. Future extensions may include:

- Integration of database storage.
- Development of a Graphical User Interface (GUI).
- Implementation of advanced sentiment analysis algorithms.

## 6. SYSTEM REQUIREMENTS

Hardware Requirements:

- Computer with a minimum of 4GB RAM.

Software Requirements:

- Java JDK.
- IDE (VS Code, Eclipse, or IntelliJ).

## 7. METHODOLOGY

The system operates through the following logical steps:

- 1.The user provides multiple feedback entries consisting of a rating and a comment.
- 2.Each entry is stored as an object of the Feedback class.
- 3.All objects are organized within an array.
- 4.The Analyzer class processes the array.
- 5.The system calculates the total feedback count and the average rating
- 6.Sentiment is categorized using switch-case logic
- 7.Final results are displayed to the user.

## . SYSTEM DESIGN

Flowchart (Textual Representation): Start → Input Feedback → Store in Array → Calculate Average → Apply Switch Case → Display Result → End

Class Diagram (Text Format):

- Feedback Class:
  - Variables: int score, String comment.
  - Methods: Constructor, Getter methods.
- Analyzer Class:
  - Methods: calculateAverage(Feedback[] feedbacks), categorizeFeedback().

## 9. IMPLEMENTATION DETAILS

The project is implemented through two primary classes:

1. Feedback Class: Represents a single entry, storing the rating and comment while providing getter methods for data access.
2. Analyzer Class: Processes multiple objects by iterating through the array to calculate the average and using switch-case logic for categorization.

Sample Input:

- Ratings: 5, 4, 4, 3, 5

Sample Output:

- Average Rating: 4.2
- Customer Sentiment: Good
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## FUTURE ENHANCEMENTS

Add dynamic user input using the Scanner class.

Implement database connectivity for data persistence.

Add a GUI interface for better user experience.

Implement keyword-based sentiment detection.

Display all individual feedback records in a summary table.

## 11. CONCLUSION

The Customer Feedback Analysis System successfully demonstrates the application of Java OOP concepts. It provides an efficient foundation for processing feedback and categorizing sentiments, serving as a base for building more advanced analytical systems.

## 12. REFERENCES

Java Official Documentation.

Online tutorials (GeeksforGeeks, W3Schools).

Class notes and study materials.